Half-Plane Shielding: Provable Directional COLREGs Compliance for Continuous-Action Safe Reinforcement Learning

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Word Count: [TBD] words + [TBD] table(s) $\times$ 250 = [TBD] words
Total Number of Pages: [TBD]

Submitted [Submission Date]
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ABSTRACT

Objectives: Autonomous surface vessels must obey the International Regulations for Preventing Collisions at Sea (COLREGs). The give-way rules are directional. A track that avoids a collision but passes on the wrong side is still a violation. This paper builds a per-step guarantee of directional compliance for continuous control commands.

Methods: The give-way clause is written as a half-plane constraint on the plane of control commands. It is intersected with the actuator limits and a one-step collision-free condition, which gives a convex safe control set. A small quadratic program projects the command of the policy onto the nearest point of that set. The projection sits inside the environment transition, so the policy learns under its own corrected commands. The data are 2,000 simulated two-vessel encounters, split into 1,400 training and 600 test cases. Nine configurations form a single-variable ladder, each trained with eight fixed seeds.

Findings: On every give-way step where the projection is feasible, directional compliance holds by construction. It does not depend on the training outcome or the seed. Steps that switch to the fallback branch are not covered. On the test set, the violation count per episode is [TBD] and the collision rate is [TBD]. The ablation gains are significant under a paired signed-rank test. The arrival rate is not claimed as a benefit.

Novelty: To the best of the authors' knowledge, this is the first projection-based safety mechanism that enforces directional COLREGs compliance on continuous control commands. The paper also states what is proved and what is not.

Practical Applications: The compliance check is independent of training. The mechanism can sit above an existing controller as a compliance layer for functional verification and liability assessment. The online cost is one small quadratic program per step, so it runs in real time without special hardware.

Keywords: COLREGs compliance, safe reinforcement learning, projection-based safety shield, continuous action space, quadratic programming, unmanned surface vehicle

4 METHODOLOGY

A note on the guarantees. Three propositions are stated in this section. Each one is placed immediately after the mechanism it applies to, rather than collected in a separate block. Each is given as a statement, its assumptions, a proof sketch, and its scope. The scope paragraph names the cases that the proposition does not cover.

4.1 COLREGs Obligations as Half-Plane Constraints

The rule state machine sorts the current encounter into six situations,

$$\rho \in \{\rho_0 \text{ conflict-free},\ \rho_1 \text{ stand-on},\ \rho_2 \text{ head-on},\ \rho_3 \text{ crossing},\ \rho_4 \text{ overtaking},\ \rho_5 \text{ emergency}\},$$

and the decision is a conjunction of three groups of conditions. Each condition is a predicate, that is, a named test on the two states that returns true or false.

The first group is the bearing sector, taken along the sidelight boundaries of the convention. The second group is the heading relation, which separates nearly anti-parallel from nearly parallel headings. The third group is collision possibility. It has two parts. The speed set of the own vessel must intersect the collision cone of the target vessel. The relative speed must also close the current center distance in time. The forward sector has a half-angle of 5° and the side sectors end at 112.5°. The overtaking band has a half-angle of 67.5°. The occupancy of the target vessel is inflated by three times its length, and the check horizon is 420 s.

Four situations carry an obligation. More than one of their predicates can hold at the same instant, so the four are tested in a fixed priority order and the first match wins. The rows below are written in that priority order, which is not the order of the indices,

$$\begin{aligned} \rho_2 \text{ head-on}: &\quad \mathrm{CP} \wedge \mathrm{Fr}(s_e, s_m) \wedge \mathrm{Ap}, \\ \rho_3 \text{ crossing}: &\quad \mathrm{CP} \wedge \mathrm{Ri}(s_e, s_m) \wedge \mathrm{Tl}, \\ \rho_4 \text{ overtaking}: &\quad \mathrm{CP} \wedge \mathrm{Be}(s_m, s_e) \wedge (v_e > v_m) \wedge \mathrm{Sd}, \\ \rho_1 \text{ stand-on}: &\quad \big[\mathrm{CP} \wedge \mathrm{Le}(s_e, s_m) \wedge \mathrm{Tr}\big] \ \vee\ \rho_4(s_m, s_e), \end{aligned} \tag{1}$$

and all other cases fall into ρ_0. Stand-on is tested last, and its definition reuses the overtaking predicate with the two vessels exchanged. Overtaking and being overtaken swap the two vessel arguments, because overtaking places the own vessel in the aft sector of the target vessel. The indices ρ_0 to ρ_5 are identifiers rather than a ranking.

Entry into a give-way situation has two parts. A give-way predicate must not hold now. It must hold at every decision step of a 60 s reaction window under constant-velocity extrapolation. The first part separates "about to hold" from "already holds", and it creates one asymmetry that Section 4.3.3 treats. The emergency situation is entered by a stronger set-based prediction with a horizon of 180 s.

Compliant control set. Each situation maps to an explicit constraint set on the command,

$$U_{\text{colregs}}(\rho) = \begin{cases} U_{\text{box}}, & \rho = \rho_0, \\ \{|a| \le \varepsilon_a\} \cap \{|\omega| \le \varepsilon_\omega\}, & \rho = \rho_1, \\ \{\omega \le -\omega_{\text{turn}}\}, & \rho \in \{\rho_2, \rho_3\}, \\ \{\omega \le -\omega_{\text{turn}}\} \text{ or } \{\omega \ge +\omega_{\text{turn}}\}, & \rho = \rho_4, \\ U_{\text{box}}, & \rho = \rho_5. \end{cases} \tag{2}$$

Head-on and crossing require a turn to starboard. Overtaking allows a turn to either side. Stand-on holds course and speed. Each set is a half-plane or an interval on the plane of control commands. A set of this shape can be entered by projection, so the commands never have to be enumerated.

Quantifying a readily apparent action. The convention asks a give-way action to be large enough to be seen in time. We quantify this as a course change of at least a given threshold within the maneuver window. With a window of 40 s and a threshold of 20°,

$$\omega_{\text{turn}} = \frac{\Delta_{\text{lt}}}{T_M} = \frac{20^\circ}{40\text{ s}} \approx 0.008727\text{ rad/s}. \tag{3}$$

This is the smallest compliant turn rate. Any turn at least this large complies, and the bound cannot be lowered.

The predicate structure and the numerical thresholds follow the temporal-logic formalization of the maritime rules (Krasowski and Althoff 2024). Three details are settled here. The inflation radius and the half-width of the speed set are stated explicitly. Entry into a give-way situation is made symmetric (Section 4.3.3). The predicate set is then mapped to a constraint set on continuous commands.

4.2 Safe Control Set and Projection

4.2.1 One-Step Collision-Free Constraint

At each decision step the applied command must keep the two hulls apart at the end of the next step. Written directly, that condition is not convex. We replace it by a convex subset of itself, built in three steps.

First, the occupancy of the target vessel after one step is over-approximated by a disk. The center follows a constant-velocity extrapolation, and the radius includes a safety margin. The own vessel is over-approximated by its circumscribed circle. The two hulls then stay apart whenever the center distance is at least the sum of the two radii, written d_{safe}.

Second, let n be the unit vector from the target vessel to the nominal end point of the own vessel, and let δ be that distance. The end point follows from the policy command clipped to the action box.

Third, the end point is expanded to first order in the command, $p_e(u) \approx p_{\text{nom}} + J(u - u_{\text{nom}})$. The end point must lie on the safe side of the separating hyperplane. This gives one linear inequality in the command:

$$\underbrace{-n^\top J}_{g^\top} u \;\le\; \underbrace{-\big(d_{\text{safe}} - \delta + n^\top J\, u_{\text{nom}}\big)}_{h}, \qquad U_{\text{cf}}(s) = \{u : g^\top u \le h\}. \tag{4}$$

A first-order expansion is neither an inner nor an outer approximation. The sign of the remainder changes with the curvature. Equation (4) alone therefore does not establish conservatism. The margin inside the safety distance is what does. With that margin the constraint acts at a center distance of 590–840 m, with a median of 714 m over this library. Hull contact needs a much smaller distance. The margin absorbs the linearization residual. The cost is a tighter constraint that rejects some commands which are in fact safe.

The linearization holds only locally. In the far field a sufficient condition is available that does not rely on it.

Proposition 1 (One-step collision freedom in the far field)**.** *Let d be the center distance between the own vessel and a single target vessel. If*

$$d \;\ge\; d_{\text{safe}} + v_{\text{obs,max}}\Delta t + \rho_{\text{ego}}, \tag{5}$$

then the two conservative circumscribed occupancies do not intersect after one decision step. Here d_{safe} is the sum of the two circumscribed radii, and ρ_{ego} is the largest displacement of the own vessel in one step.

Assumptions. A single target vessel. The speed of the target vessel is at most 9.5 m/s. That bound is not implied by the dynamics and is stated separately; the largest value measured over the 2,000 scenarios of the library is 7.10 m/s. The target vessel holds its speed within the step.

Proof sketch. By the triangle inequality, the center distance after the step is at least d minus the one-step displacement bound of each vessel. It is therefore at least d_{safe}.
Scope. The triangle inequality does not depend on direction, so a head-on encounter is the worst case. With the constants of this library the threshold is about 764 m. The statement covers one step. It does not extend to the steps after it, and it does not cover the near field.

4.2.2 Projection as a Quadratic Program

The safe control set is the intersection of three sets,

$$U_{\text{safe}}(s,\rho) = U_{\text{box}} \cap U_{\text{colregs}}(\rho) \cap U_{\text{cf}}(s), \tag{6}$$

and it is a convex polygon in $\mathbb{R}^2$. The applied command is the Euclidean projection of the policy command onto that set,

$$u_{\text{safe}} = \Pi_{U_{\text{safe}}}(u_{\text{policy}}) = \arg\min_{u \in U_{\text{safe}}} \|u - u_{\text{policy}}\|_2^2. \tag{7}$$

The feasible set is an intersection of half-planes, so it can be written as $U_{\text{safe}} = \{u : Au \leq b\}$. The first four rows are the action box, the fifth is the compliant-direction half-plane, and the sixth is the collision-free constraint of Equation (4). The program has two variables and at most six constraints. It is small enough to solve online at every decision step.

Proposition 2 (Directional compliance on every give-way step)**.** *Consider any decision step that the rule state machine assigns to a give-way situation and on which the projection is feasible. The applied command* u_{safe} *then lies in the compliant-direction half-plane.*

Assumptions. A single target vessel, and a correct decision by the state machine. The second assumption inherits one conservative bias of the collision-possibility predicate. That predicate uses the center distance, and it can miss a trailing encounter in which the two speeds are close.
Proof sketch. The compliant half-plane enters the quadratic program as a hard constraint. Any feasible point satisfies it, and the projection returns a feasible point. The statement holds by construction. It does not depend on the training outcome or on the random seed.
Scope. Emergency situations are not covered, because their branch bypasses the compliance constraint by design. Steps on which the safe control set is empty and the fallback is used are not covered either.

4.2.3 Fallback and the Unavoidable-Collision Criterion

When the safe control set is empty, the projection has no solution. A substitute rule then supplies the command. We call it the fallback, and it carries no guarantee. The fallback branches on the encounter situation. It does not try a fixed list of options in order. The two branches below are mutually exclusive.
Emergency branch. An emergency situation takes a separate path. No collision-free constraint is built, and Equation (7) is not solved. A stateful geometric controller supplies the command instead, and its lifetime is one emergency event. It has three modes. A near head-on approach ahead is handled by tracking a fixed lateral point fixed at the onset of the emergency. A target vessel astern that can be cleared by acceleration is handled by a straight-line escape inside the reaction window. Every other case tracks a point that moves with the target vessel. One position-tracking law converts all three into a control command. This controller offers no guarantee. Its purpose is a defined behavior once no compliant feasible command remains.
Other situations. Outside an emergency situation the fallback is used only when the safe control set is empty. A degenerate case is tested first. If the linearization gives no usable separation direction, the command is projected onto the action box alone. Otherwise the direction constraint is relaxed, and the collision risk is minimized if that also fails. The last two branches solve on the actuator limits and drop the compliance constraint as a whole. The turn rate can then leave the policy action box, so saturation is reported against each box separately.

These branches are needed in the implementation, and on the scenario library used here they almost never fire. If the own vessel holds course and speed, the median closest center distance to the target vessel is about 1,300 m. The safe control set is therefore rarely empty. Purpose-built conflict cases are constructed so that the branches are exercised, and their outcome is reported with the shield measurements.

The emergency branch offers no guarantee. It can still decide whether the situation was already beyond recovery.

Proposition 3 (Conservative criterion for an unavoidable collision)**.** *Consider a single target vessel at constant velocity. Let $R_{\text{box}}(t)$ be an over-approximation of the reachable centers of the own vessel, and let $O(t)$ be the hull rectangle of the target vessel. If*

$$\exists t^* \in [0,T]: \quad R_{\text{box}}(t^*) \subseteq \big(O(t^*) \oplus \text{disk}(r_{\text{insc}})\big), \tag{8}$$

then the collision is unavoidable, and every feasible control sequence meets that target vessel at t^*.

Assumptions. A single target vessel at constant velocity. The hull size of the target vessel must not be overstated, since a beam larger than the true value makes the test unreliable. The horizon is 120 s, evaluated at 241 equally spaced instants.
Proof sketch. The rate of change of the velocity vector is bounded. A double integration gives the two semi-axes of R_{box}, which therefore over-approximates the true reachable centers. The inscribed circle under-approximates the hull. Once every reachable center lies inside the inflated occupancy of the target vessel, no control command can avoid contact.
Scope. The criterion is one-sided. It can miss a case, and it never raises a false alarm. A negative result therefore does not mean that the collision is avoidable. The criterion also acts late rather than early, and maneuvering target vessels are not covered. On 20,000 random scenarios and 735 selected scenarios it produced no false positive. After the envelope was hardened, 1,500 further fuzz cases produced none either.

4.3 Shielded Policy Optimization

The shield is not applied after the policy as a post hoc correction. The projection of Equation (7) is placed inside the environment transition. The policy emits u_{policy}, the environment applies u_{safe}, and it returns the successor state and the reward. The policy therefore faces the composed transition

$$s_{t+1} \sim P\big(\cdot \;\big|\; s_t,\, \Pi_{U_{\text{safe}}(s_t,\rho_t)}(u_t)\big), \qquad u_t \sim \pi(\cdot \mid s_t), \tag{9}$$

rather than the $P(\cdot \mid s_t, u_t)$ of a post hoc filter. The policy learns on the transition it will actually meet, so it adapts to the projection during training instead of meeting an unfamiliar filter afterwards.
Learning algorithm. All configurations are trained with proximal policy optimization (Schulman et al. 2017). The continuous-action configurations use its standard form. The discrete-action baselines use its maskable variant, because action masking is what makes a discrete shield possible. Both action spaces therefore share one policy-gradient algorithm, which removes a source of difference between them. The masking itself remains a difference, and it is intrinsic to the discrete method rather than a choice made here.

4.3.1 Reward Function

The reward is a sum of five terms,

$$r = r_{\text{sparse}} + r_{\text{goal}} + r_{\text{velocity}} + r_{\text{deviate}} + r_{\text{colregs}}. \tag{10}$$

A sparse terminal term scores arrival, timeout, stalling, and collision, and it charges a fixed penalty for every step that enters emergency control. A dense approach term rewards the change in distance to the goal over one step. A speed term penalizes speed outside the cruise band [2.5, 8.0] m/s. A cross-track term penalizes lateral deviation from the start-to-goal line and saturates at 2,000 m. A soft compliance term decays with distance. It is active only in the reward-shaping baseline and in the discrete configurations. Every continuous configuration here sets it to zero, because compliance is carried by the hard constraint of Equation (2).

The five terms are taken from (Krasowski and Althoff 2024), with three deviations that we declare. The approach term uses the difference of distances between two steps. The soft compliance term is rescaled, because its original calibration is set at a scale about fifty times smaller than the scenarios used here. One clause of the emergency-release test is read by its physical meaning rather than by its literal sign, since the two disagree.

4.3.2 Bounded Action Distribution

A continuous policy usually emits an unbounded Gaussian and clips the sample to the action box before it reaches the environment. That combination conflicts with Equation (7), for two reasons that both come from the clipping. First, common implementations feed the clipped action to the environment but the unclipped action to the policy gradient. Once the mean leaves the box, moving further out changes no reward, the gradient reaches a plateau, and any penalty on action smoothness loses its grip. Second, the differential entropy of a Gaussian has no upper bound, while the entropy coefficient is a constant. The standard deviation therefore has a one-way channel to grow. Its growth pushes still more samples outside the box.

We use a Beta distribution with support on a closed interval. Actions can then never leave the box, clipping becomes the identity map, both channels close, the smoothness penalty is differentiable everywhere, and the entropy is bounded above. The network emits two reals per control axis, and

$$\alpha = \mathrm{softplus}(\tilde{\alpha}) + 1, \quad \beta = \mathrm{softplus}(\tilde{\beta}) + 1, \qquad u = u_{\mathrm{lo}} + (u_{\mathrm{hi}} - u_{\mathrm{lo}})\,z, \quad z \sim \mathrm{Beta}(\alpha, \beta) \tag{11}$$

gives the shape parameters and the applied action. Evaluation uses the mode instead of a sample. The offset of one keeps both shape parameters above unity, and that constraint cannot be dropped. Below unity the Beta density is U-shaped and diverges at the two ends, and the mode then sits at an endpoint. Because evaluation always takes the mode, such a policy would silently corrupt every reported number, and the aggregate metrics would give no sign of it.

4.3.3 Symmetric Entry into Give-Way Situations

In the state machine, entry into a give-way situation is not symmetric. From the conflict-free situation it is triggered only by the persistence condition. From the stand-on situation a separate immediate condition also applies. We add the symmetric branch. When the persistence condition does not fire, the immediate condition is checked, and the give-way situation is entered if it already holds. The obligation is then recognized at the same moment on both routes. The change alters the command that the shield applies, so training and evaluation must use the same setting.

A purpose-built case shows why the branch is needed. Suppose an encounter already satisfies a give-way predicate at the initial instant. The clause “does not hold now” can then never be satisfied. The persistence condition never fires, and the encounter is never recognized as a give-way situation. On a set of such adversarial cases the persistence route alone gives 0 give-way decision steps. With the immediate branch it gives 201. These cases are synthetic and illustrate the mechanism, so they are reported apart from the main results.

4.3.4 Training Setup

Apart from the entropy coefficient, which is set to 0.01, and the bounded action distribution, every hyperparameter takes the default of the implementation library and is left untuned. The libraries are Stable-Baselines3 2.3.2 (Raffin et al. 2021) and sb3-contrib 2.3.0, and the advantage is computed by generalized advantage estimation (Schulman et al. 2016). The discrete and continuous configurations agree on every one of these values, so the hyperparameters are not a source of difference between them. Running normalization of the observation and the reward is not optional. Without it the large-scale approach term in Equation (10) drives the policy to a degenerate solution that holds station.

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